Medical Device Analytics Dashboard Project Report

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Installation coverage 1 January 2023 to 31 December 2025
Report date 5 November 2026

Executive summary

This Excel project brings device inventory, recorded revenue, service costs, downtime, compliance and warranty information into one analytical workbook. It covers 3,000 unique device records across 18 facilities and 18 cities. The Quantity field represents 7,000 units, so record counts and unit counts answer different questions.

Key measure	Verified result
Device records	3,000
Recorded quantity	7,000 units
Net revenue	£134,235,725.43
Total service cost	£14,969,761.99
Recorded downtime	14,889.4 hours
Compliant records	2,470 of 3,000 / 82.33%
High or Critical records	293 of 3,000 / 9.77%
Average satisfaction score	90.59

Source: Dashboard!A5, E5, I5, M5, Q5, U5; Cleaned_Data!S2:S3001, AB2:AB3001. Figures use all records and the saved installation-date window.

The most concentrated service-cost opportunity is the 368 records labelled High Cost. They represent 12.27% of records but account for £6,049,442.61, or 40.41%, of service expense. Radiology has the highest departmental service cost and downtime. These findings support targeted review of expensive service cases and departments with substantial recorded disruption.

The workbook contains 13 actual PivotTables and 12 dashboard charts. Its six saved headline values reconcile with the underlying data. Before recurring use, the priority improvements are consistent filtering, complete chart coverage, a documented reporting date and reproducible classification rules. The report distinguishes the implemented Excel features from the transformation steps described in the workbook.

Source: Cleaned_Data!K2:K3001, Z2:Z3001, AQ2:AQ3001; Pivot_Tables!A29:B42, D51:E64; workbook object and formula inspection.

1 Project purpose and scope

The project addresses the needs of a UK medical device operations team that must monitor installed equipment, service expenditure, recorded revenue and operational follow-up across facilities. The analysis is intended to help an analyst explain where activity is concentrated and identify records that need closer investigation.

Analytical objectives

The objectives are to organise device records into a consistent analytical table, summarise financial and operational measures, compare facilities and departments, expose compliance and warranty follow-up groups, and provide filters for focused dashboard review. Recommendations concern the recorded dataset and the reporting process.

Dataset characteristic	Coverage
Raw and cleaned records	3,000 in each table
Fields	31 original fields and 16 added fields
Device coverage	10 categories and 42 device types
Organisational coverage	18 facilities, 18 cities and 12 regions
Service coverage	12 departments and 20 named technicians
Client and supplier coverage	5 client types and 12 manufacturers

Source: Raw_Data!A1:AE3001; Cleaned_Data!A1:AU3001. Counts are distinct observed values.

Workbook structure

Worksheet	Role in the project
Project_Overview	Problem statement, scope and deliverables
Raw_Data	Original 3,000-record table
Cleaned_Data	Analysis table with financial, date and status enrichment
Pivot_Tables	13 grouped summaries and grand totals
Dashboard	Six KPI cells, seven dropdowns, two dates and 12 charts
Power_Query_Steps	12 documented preparation and refresh steps
Data_Dictionary	Descriptions of all 47 analysis fields
Business_Insights	Eight questions with summary observations
Lists	Hidden dropdown option lists
Chart_Data	Hidden formula results used by dashboard charts

Each row has a unique Device_ID, but Quantity can exceed one. Monetary totals already incorporate the amounts recorded for that row and are not multiplied by Quantity again. The source-system provenance is not recorded in the file, so these findings describe the project dataset rather than verified activity at a named healthcare organisation.

Source: Project_Overview!B7:B13; Data_Dictionary!A5:B51; worksheet, table and package inspection.

2 Data preparation and validation

Raw_Data contains 3,000 records and 31 fields. Cleaned_Data retains the same 3,000 Device_IDs and all original field values, and adds 16 analytical fields. The demonstrated change is enrichment of an already complete table. A raw-to-clean comparison does not show removed duplicates, replaced blanks or corrected source values.

Documented preparation workflow

Power_Query_Steps describes importing data, checking duplicates and invalid values, standardising text, assigning data types, creating conditional and custom columns, loading the transformed table and refreshing summaries. These steps provide a useful workflow specification. The file contains stored raw and cleaned tables without an embedded executable Power Query connection or query definition.

Source: Raw_Data!A1:AE3001; Cleaned_Data!A1:AU3001; Power_Query_Steps!A5:C16; workbook connection and query-part inspection.

Observed data checks

Check	Observed result
Unique Device_ID values	3,000 in both tables; no duplicate IDs
Original-field preservation	All 31 original fields match by Device_ID
Missing cells	None across the populated raw or cleaned fields
Negative numeric values	None in the checked monetary, quantity, downtime, ticket, age and score fields
Date order	No installation before purchase; no warranty end before installation
Row arithmetic	Gross, net, service cost and combined device value reconcile
Date attributes	Installation years agree with installation dates
Serial-number uniqueness	One repeated serial value occurs in two records
Saved formula errors	No stored Excel error cells found

Source: Independent comparison and recomputation of Raw_Data and Cleaned_Data; stored-cell error scan across all worksheets.

Serial number follow up

Serial number MD23381377 appears in records MD-00325 and MD-02842 at Cleaned_Data!G326 and G2843. The two records have different manufacturers and models. This should be checked against the intended identifier rule before changing either record. A manufacturer-and-serial combination may be more meaningful than a serial number alone, depending on the source system.

The preparation record should retain the input row count, output row count, rejected rows and reasons, affected fields and refresh date. For this version, the accurate outcome is zero removed records, zero changes to original values and 16 added fields.

3 Metric definitions and calculation methods

All headline values use the saved All selections and installation dates from 1 January 2023 to 31 December 2025. Counts use records, financial measures sum the recorded GBP amounts, and the satisfaction result is an unweighted mean across records.

Measure	Calculation and meaning	Result
Device records	Count qualifying Device_ID rows	3,000
Recorded quantity	Sum Quantity	7,000
Gross revenue	Unit_Price_GBP × Quantity, summed	£140,348,395.52
Discounts	Sum Discount_GBP	£6,112,670.09
Net revenue	Gross_Revenue_GBP − Discount_GBP	£134,235,725.43
Service cost	Maintenance + parts + training	£14,969,761.99
Device expense field	Gross revenue + service cost	£155,318,157.51
Compliance rate	Compliant records ÷ qualifying records	82.33%
High or Critical share	(170 High + 123 Critical) ÷ 3,000	9.77%
Satisfaction	Mean of Customer_Satisfaction_Score	90.59
Downtime	Sum Downtime_Hours	14,889.4 hours
Service tickets	Sum Service_Tickets	2,798

Source: Cleaned_Data!R2:AC3001, AF2:AF3001, AH2:AJ3001; Dashboard!A5:U5; Data_Dictionary!A22:B40.

Interpretation of derived measures

Service cost equals 11.15% of recorded net revenue, calculated from £14,969,761.99 divided by £134,235,725.43. This ratio compares two dataset totals. It does not establish profitability because acquisition cost, overhead and accounting recognition dates are not separately identified. The field Total_Device_Expense_GBP explicitly adds gross revenue to service cost, so its £155,318,157.51 total should retain that stated definition.

The satisfaction field is a numerical score with no documented scale in the dictionary. The report therefore uses “90.59 score”, without converting it into a percentage. Similarly, total downtime has no defined observation-period or available-hours denominator, so an uptime percentage cannot be calculated from the supplied fields.

Installation year, month and quarter come from Installation_Date. Risk, warranty, age bands, renewal priority and operational flags are stored classifications. The dictionary describes their purpose but does not fully specify the cutoffs, rule order or dated calculation basis needed to reproduce them.

Source: Data_Dictionary!A31:B51; Cleaned_Data!AG2:AU3001. Ratio and aggregation notes are calculated for this report.

4 Dashboard and PivotTable implementation

The dashboard combines six formula-based KPIs with 12 standard Excel charts. Chart values come from 157 formulas on the hidden Chart_Data sheet. The 13 PivotTables form a separate summary layer, all based on the cleaned dataset. The charts are ordinary charts linked to formula results rather than PivotCharts.

Dashboard chart	Type	Measure and scope
Total Devices by Facility	Bar	Record counts for 10 fixed facilities
Department Service Cost	Column	Service cost across 12 departments
City Device Distribution	Pie	Record counts for 10 fixed cities
Payment Status Revenue	Doughnut	Net revenue by four invoice statuses
Monthly Service Trends	Two lines	Record count and cost by installation month
Device Status Mix	Doughnut	Four stored operational statuses
Contract Type Service Expense	Bar	Service cost by six contract types
Risk Level Distribution	Pie	Four stored risk groups
Warranty Status	Doughnut	Three stored warranty groups
Technician Service Tickets	Column	Ticket totals for 10 fixed technicians
Average Satisfaction by Facility	Bar	Mean score for 10 fixed facilities
Downtime by Department	Column	Hours across 12 departments

Source: Dashboard chart definitions; Chart_Data!A1:C167. Detailed source ranges and PivotTable references appear in Section 15.

How to use the controls

Start with All in the seven dropdowns. Select device category in B8, facility in D8, department in F8, region in H8, contract type in J8, payment status in L8 and risk in N8. Set the installation-date window in B10 and D10. Change one selection at a time and check the record count before interpreting rates or averages.

These controls are data-validation dropdowns and date cells. They are separate from native slicers, timelines and PivotTable filters. Reset the dropdowns to All and the dates to the full window when comparing the report totals. The Pivot_Tables sheet can be used to inspect complete category summaries and the underlying field arrangements.

The Risk filter omission in U5 and the fixed category lists require care during filtered review. An empty chart does not always mean an empty selection, and zero satisfaction or compliance can represent a no-match formula fallback. Section 11 explains these implementation limits.

Source: Dashboard!A5:U10; Lists!A1:G20; Pivot_Tables!A5:E158; formula and validation definitions.

5 Revenue and commercial concentration

The dataset records £140,348,395.52 of gross revenue and £6,112,670.09 of discounts, resulting in £134,235,725.43 of net revenue. Discounts represent 4.36% of gross revenue. These totals describe the records selected by installation date, rather than an independently defined financial reporting period.

Net revenue by device category

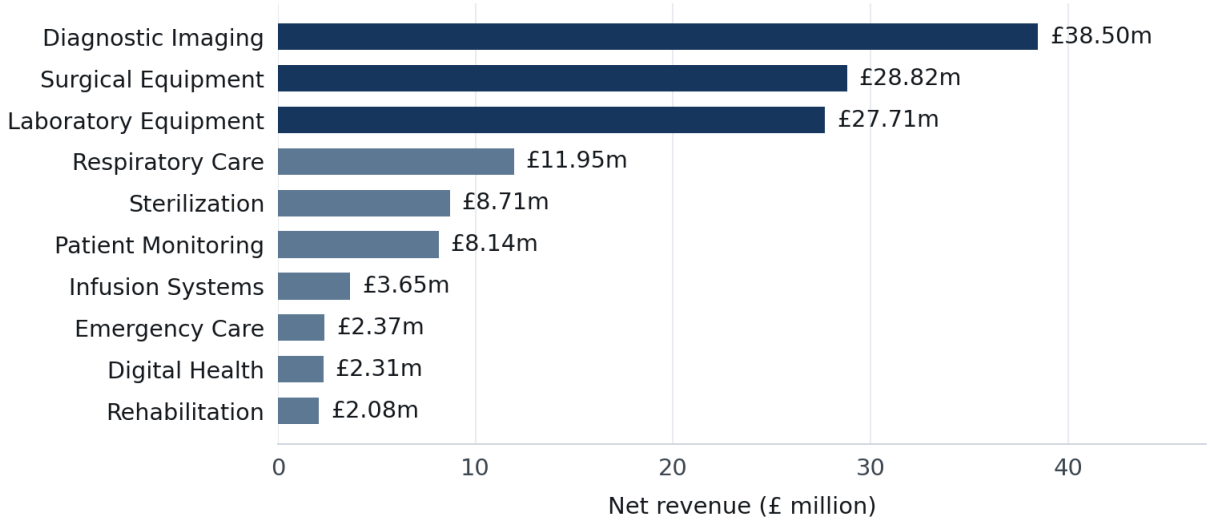


Figure 1 Recorded net revenue by device category

Source: Cleaned_Data!C2:C3001 and V2:V3001. Categories ranked by independently aggregated net revenue.

Diagnostic Imaging contributes £38,498,026.16, followed by Surgical Equipment at £28,824,157.33 and Laboratory Equipment at £27,707,052.64. Together these three categories represent 70.79% of recorded net revenue. Their commercial importance supports reviewing service commitments alongside the value of the installed equipment.

Revenue by coverage category

Coverage label	Net revenue	Share
Service Contract	£41,528,974.44	30.94%
Manufacturer Warranty	£33,327,194.71	24.83%
NHS Framework	£32,148,407.30	23.95%
Private Cover	£14,040,144.60	10.46%
Self Funded	£13,191,004.38	9.83%

Source: Pivot_Tables!D31:E37; Cleaned_Data!Q2:Q3001, V2:V3001.

Service Contract is the largest label within Insurance_Coverage at £41.53 million. Insurance_Coverage and Contract_Type are separate fields and should retain their distinct meanings. The table measures recorded revenue associated with each label; it does not quantify reimbursement received or warranty recoveries.

6 Service costs and recorded downtime

Total service expense is £14,969,761.99. Maintenance contributes £9,266,735.39 (61.90%), parts £2,831,023.38 (18.91%) and training £2,872,003.22 (19.19%). Average service expense is £4,989.92 per record. These are record-level totals and are not weighted again by Quantity.

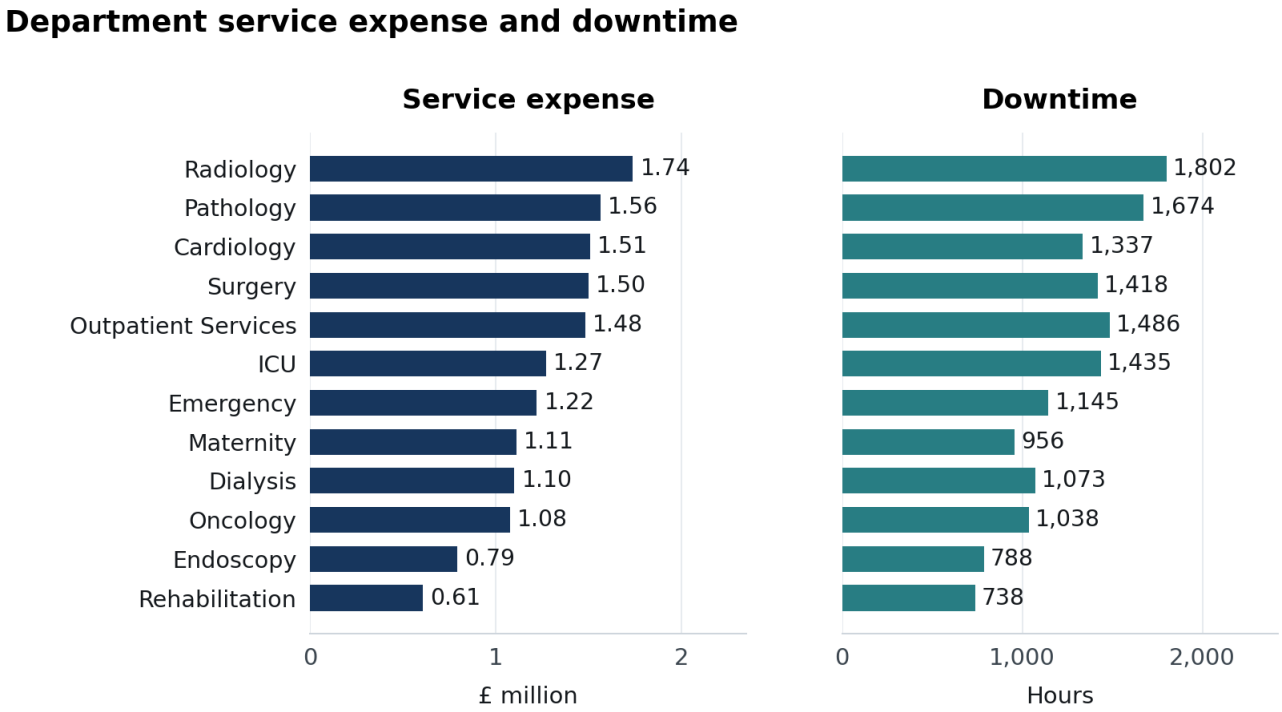


Figure 2 Service expense and recorded downtime by department

Source: Pivot_Tables!A29:B42 and D51:E64; Cleaned_Data!W2:Z3001. Percentages calculated from total service cost.

Radiology leads both measures with £1,738,853.60 of service cost and 1,801.8 downtime hours. Pathology follows on both totals. Radiology, Pathology and Outpatient Services together contribute 4,961.8 hours, or 33.32%, of all recorded downtime. Their underlying service cases warrant a focused review of recurrent faults, parts delays and maintenance requirements.

Concentration within service records

Stored group	Records	Relevant total	Share of total
High Cost	368	£6,049,442.61 service cost	40.41% of service cost
High or Critical risk	293	3,385.4 downtime hours	22.74% of downtime
Long Downtime	16	1,405.2 downtime hours	9.44% of downtime

Source: Cleaned_Data!Z2:Z3001, AB2:AB3001, AH2:AH3001, AQ2:AR3001. Groups overlap and should not be added together.

Absolute totals should be considered alongside exposure and device mix. Maternity has the highest departmental service cost per record at £5,853.10, despite ranking below several departments on total cost. This highlights why a high total alone does not prove poor efficiency or avoidable spending.

7 Compliance and operational risk

The workbook labels 2,470 records Compliant, 418 Review Due and 112 Non-Compliant. The compliance KPI is the proportion labelled Compliant: 82.33%. Review Due and Non-Compliant describe different follow-up groups and should remain distinct when presenting the backlog.

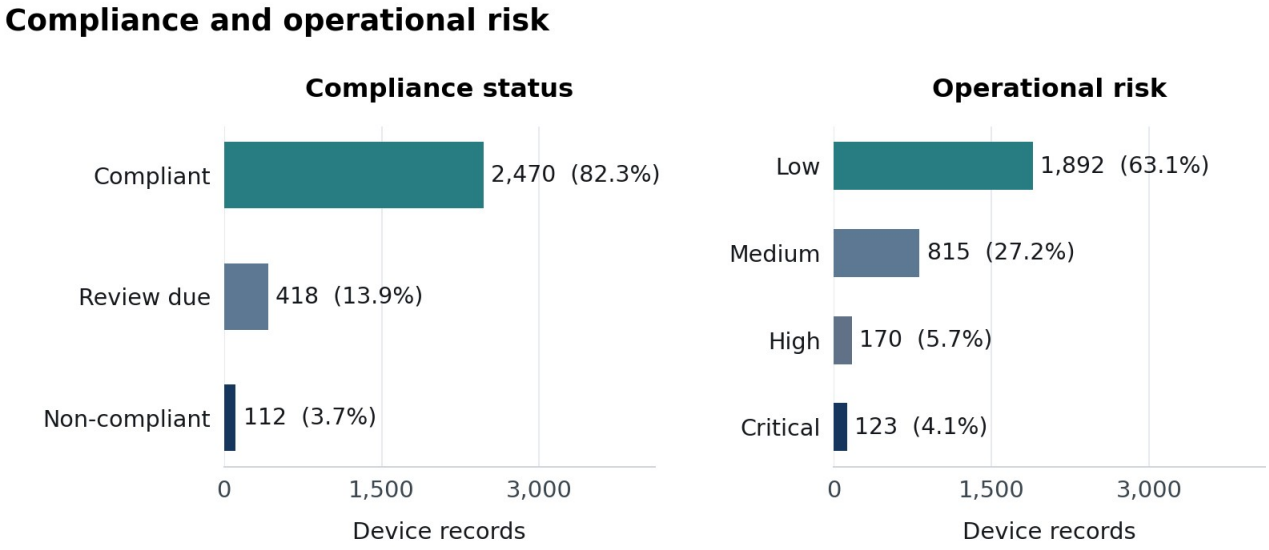


Figure 3 Stored compliance and risk classifications across all 3,000 records

Source: Pivot_Tables!D69:E73 and D78:E83; Cleaned_Data!AF2:AF3001, AH2:AH3001. Shares use all 3,000 records.

Overlap and prioritisation

There are 170 High and 123 Critical records, giving 293 combined records, or 9.77%. All 112 Non-Compliant records also belong to the Critical group. Adding these populations would therefore double-count records. A useful worklist should retain one Device_ID per case with separate risk and compliance fields.

The combined High and Critical group accounts for £2,094,614.56 of service cost and 3,385.4 downtime hours. Its shares are 13.99% of service cost and 22.74% of downtime, compared with 9.77% of records. This is a descriptive association because downtime and compliance contribute to the stated risk definition.

Follow-up population	Records	Interpretation
Non-Compliant	112	Every record is also labelled Critical
Review Due	418	Separate review backlog; does not imply non-compliance
Review Due and Stable	377	Not surfaced by the Needs Review flag alone
Needs Review	293	Matches the combined High and Critical population

Source: Cross-tabulation of Cleaned_Data!AF2:AF3001, AH2:AH3001 and AU2:AU3001.

The operational review should therefore include the compliance-status backlog directly, as well as the risk and improvement flags. The workbook labels support case selection; the underlying review evidence and classification rules are required to substantiate the status of each record.

8 Warranty renewal and equipment age

Warranty and age results below are the classifications stored in the workbook. They are not recalculated to the report date. A controlled analysis date is needed to keep time-dependent fields consistent during future refreshes.

Stored warranty status	Records	Share	Service cost
In Warranty	1543	51.43%	£7,456,423.46
Extended Warranty	611	20.37%	£2,924,916.13
Out of Warranty	846	28.20%	£4,588,422.40

Source: Cleaned_Data!AG2:AG3001 and Z2:Z3001; Data_Dictionary!A37:B37.

The 846 Out of Warranty records represent 28.20% of all records and account for £4,588,422.40 of service costs. Their mean service cost is £5,423.67 per record. The 611 Extended Warranty records should be interpreted using the underlying contract terms rather than treated automatically as expired coverage.

Renewal priorities

Stored renewal priority	Records	Share
Immediate	91	3.03%
High	1049	34.97%
Medium	497	16.57%
Normal	1363	45.43%

Source: Cleaned_Data!AS2:AS3001. Priority cutoffs are not fully specified in the workbook.

All 91 Immediate renewal records are Out of Warranty and also High or Critical. They provide a defined first group for contract-status verification. The 1,049 High-priority records form a wider follow-up population. These groups should be reviewed against current warranty dates and service-contract information before taking action.

Stored age profile

Stored age band	Records	Share
0-1 Year	518	17.27%
1-3 Years	2015	67.17%
3-5 Years	467	15.57%

Source: Pivot_Tables!D42:E46; Cleaned_Data!AO2:AP3001, AT2:AT3001.

The largest band is 1-3 Years with 2,015 records, or 67.17%. All 3,000 records have Replacement_Recommended set to No. This uniform flag provides no ranking for replacement planning. Any future replacement analysis needs a documented rule and relevant inputs such as remaining useful life, repair history, utilisation and replacement cost.

9 Inventory distribution and installation cohorts

Queensbrook Diagnostics has the largest record count at 192, equal to 6.40% of the dataset. Every city in this workbook corresponds to one facility, so facility and city counts describe the same distribution under different labels.

Facility	City	Records	Share
Queensbrook Diagnostics	Edinburgh	192	6.40%
Orchard Lane Hospital	Nottingham	187	6.23%
Riverstone Clinical Centre	Leicester	182	6.07%
Brookfield Cardiac Centre	York	181	6.03%
Northmere General Hospital	Manchester	174	5.80%

Source: Pivot_Tables!A5:B24 and A71:B90; Cleaned_Data!H2:I3001. Full dataset used for percentage denominators.

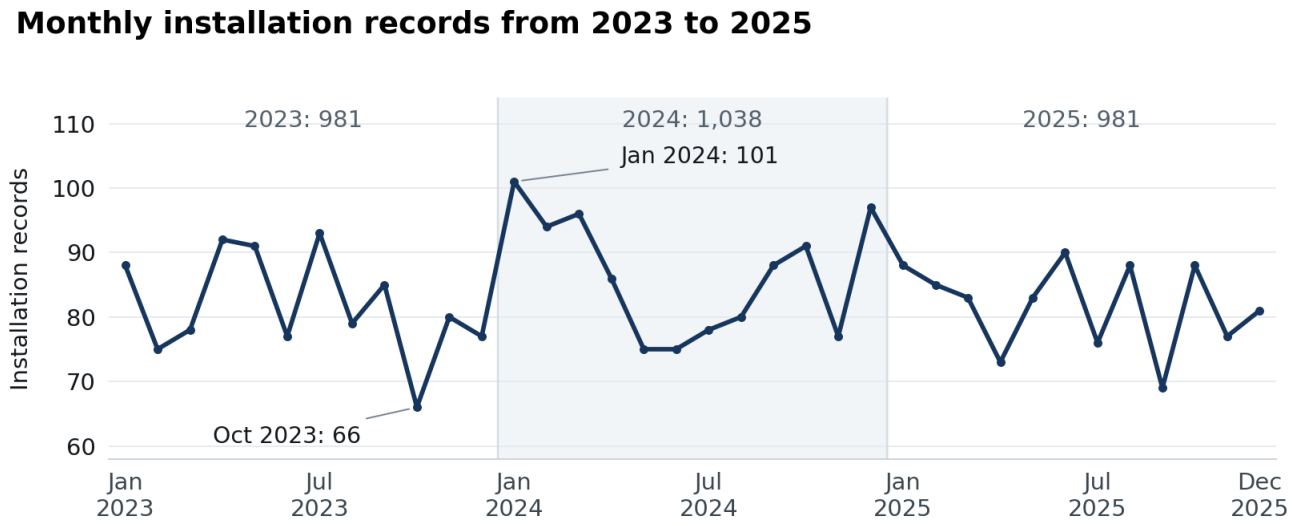


Figure 4 Installation records by month from January 2023 to December 2025

Source: Pivot_Tables!A121:B158; Cleaned_Data!N2:N3001 and AM2:AM3001.

The dataset contains 981 installation records in 2023, 1,038 in 2024 and 981 in 2025. This is a 5.81% increase in record count in 2024 followed by a 5.49% decrease in 2025. January 2024 is the highest month at 101 records, while October 2023 is the lowest at 66.

Installation cohort	Records	Associated service cost
2023	981	£5,969,346.49
2024	1038	£4,978,662.92
2025	981	£4,021,752.58

The service figures are costs associated with devices installed in each year. The file has no service-event date, so they cannot establish when that servicing occurred. Comparisons between cohorts also require care because older installations may have had more time to accumulate service activity.

Source: Cleaned_Data!AK2:AK3001, Z2:Z3001; Data_Dictionary!A41:B44.

10 Payments service tickets and satisfaction

Recorded revenue by invoice status

Invoice status	Records	Associated net revenue	Revenue share
Paid	1978	£90,259,895.89	67.24%
Pending	512	£22,118,293.53	16.48%
Partially Paid	321	£13,090,397.08	9.75%
Overdue	189	£8,767,138.93	6.53%

Source: Pivot_Tables!A95:B100; Cleaned_Data!AI2:AI3001 and V2:V3001.

The 189 records marked Overdue have £8,767,138.93 of associated net revenue. Records marked Pending, Partially Paid or Overdue together have £43,975,829.54 of associated value. This is not an exact outstanding receivables balance because the file does not include amounts already collected, remaining balances or payment due dates.

Service activity and workload

The dataset contains 2,798 service tickets. Surgical Equipment has the largest category total at 481 tickets, followed by Laboratory Equipment at 440. Laboratory Equipment has the highest ticket count per record at 1.31 and the highest category downtime at 2,637.5 hours. These measures help identify cases for review but do not by themselves measure failure rates without consistent exposure information.

Technician	Recorded service tickets
Aarav Desai	175
Arjun Patel	164
Rohan Gill	151
Karan Joshi	150
Dev Malhotra	147

Source: Pivot_Tables!A105:B116 and D5:E26; Cleaned_Data!C2:C3001, AB2:AD3001.

Aarav Desai has the highest assigned total at 175 tickets, followed by Arjun Patel at 164. These are activity totals associated with the named technician. Resolution time, working hours, case complexity and completion quality are not measured, so the ranking should support workload review rather than individual performance conclusions.

Customer satisfaction

Average satisfaction is 90.59. Horizon Day Surgery Centre has the highest facility mean at 91.26 across 161 records. Avalon Biomedical Centre has the lowest at 90.08 across 166 records. The spread between the highest and lowest facility averages is only 1.18 score points. Interpretation should retain the sample counts and the absence of a documented score scale or response methodology.

Source: Pivot_Tables!A47:B66; Cleaned_Data!H2:H3001 and AJ2:AJ3001.

11 Verification and implementation limits

Independent aggregation confirms all six saved KPI values, 176 PivotTable category and total values, and 157 chart-source values. Chart caches agree with their source cells, and no stored Excel error cells were found. This establishes consistency of the saved baseline. Interactive behaviour was assessed through formulas and object definitions rather than a Microsoft Excel desktop session.

Filter consistency

Dashboard!U5 calculates High and Critical counts without referencing the Risk selection in N8. Its unfiltered result of 293 is correct. With the other controls unchanged, selecting Low implies 1,892 records in the main count while U5 remains 293. The KPI should apply the same risk condition as the rest of the view or be labelled explicitly as a comparison across all risk groups.

Chart coverage and percentages

Four charts retain fixed lists of ten facilities, cities or technicians. Their labels do not dynamically re-rank when filters change. The city pie includes 1,749 records, or 58.30% of the full dataset. Eight omitted cities contain 1,251 records. Edinburgh is 6.40% of all records but 10.98% of the plotted subset. A complete bar chart or an explicit Other cities category would prevent this denominator ambiguity.

Source: Dashboard!U5 and N8; Chart_Data!A3:B12, A33:B42, A128:B137, A142:B151; Pivot_Tables!A71:B90.

Refresh and reproducibility

The PivotTable source and dashboard formulas stop at row 3001. Category lists are also fixed. New rows or new categories therefore require explicit range maintenance unless the implementation is changed to use expanding table references and updated lists. The narrative Power Query steps do not provide a runnable transformation in the supplied file.

Derived statuses are stored values. Numerical thresholds, the order of classification rules and an explicit analysis date should be documented before repeating the calculation. Age and warranty labels should be refreshed together using the same date. The report preserves the supplied classifications rather than presenting them as current on 5 November 2026.

Source: Cleaned_Data!A1:AU3001; Dashboard and Chart_Data formula ranges; PivotTable cache source; Power_Query_Steps! A5:C16; Data_Dictionary!A37:B51.

No matching records and date order

The satisfaction and compliance formulas return zero when no records qualify because they use IFERROR. An undefined average or rate should display a no-records result instead. The two date inputs have individual validation but no check that the start date precedes the end date. Both conditions should be visible to a dashboard user.

Source: Dashboard!M5, Q5, B10 and D10; data-validation definitions.

12 Recommendations and project conclusion

First establish consistent reporting behaviour

Apply the Risk filter consistently to the High/Critical KPI. Replace fixed top-ten lists with full category coverage or a ranking that updates after filtering. Label the monthly chart as an installation-cohort comparison. Present “device records” and “recorded quantity” as separate KPIs, and use a no-matching-records state for undefined rates and averages. These changes address the interpretation problems identified in Sections 3, 9 and 11.

Make preparation and refresh repeatable

Implement the documented transformation workflow as a retained query or an equally auditable calculation process. Record the analysis date, all classification cutoffs and the order in which rules apply. Use expanding source references, a refresh log and reconciliations for row count, revenue and service cost. Retain the raw input and keep rejected records with their rejection reasons when future data requires cleaning.

Focus operational review on defined populations

Begin service-cost review with the 368 High Cost records, which contribute 40.41% of service expense. Investigate the 16 Long Downtime records that contribute 9.44% of downtime, and examine the Radiology, Pathology and Outpatient Services cases behind 33.32% of downtime. Compare case details and device mix before setting reduction targets. The dataset does not establish an achievable saving.

Maintain separate worklists for the 112 Non-Compliant and 418 Review Due records, deduplicated by Device_ID when combined with risk groups. Verify current coverage for the 91 Immediate renewal records. For the 189 Overdue invoice records, add remaining balance, amount received and due date before reporting a receivables position.

Add the fields needed for stronger analysis

Service-event dates and ticket resolution dates would enable true service-period trends. Available operating hours and utilisation would support exposure-adjusted downtime comparisons. Invoice balances would support collections reporting. A documented satisfaction scale and response method would strengthen comparisons. A defined source-system owner and refresh date would clarify provenance and accountability.

Project outcome

The project demonstrates a structured Excel data model, 16-field enrichment, real PivotTables, conditional aggregation formulas, validation dropdowns and a multi-chart dashboard. It turns 3,000 records into traceable views of financial value, service burden and follow-up groups. The saved summaries reconcile, and the strongest findings identify concentrated service spending and downtime. With consistent filtering and reproducible refresh rules, the same structure can support recurring operational reporting.

Source: Findings in Sections 2-11. Supporting fields and PivotTable ranges are indexed in Section 15.

13 Data dictionary for original fields

The following 31 fields are present in both Raw_Data and Cleaned_Data. Column letters refer to Cleaned_Data.

Field	Col	Definition
Device_ID	A	Unique identifier for the record
Device_Name	B	Device type and model description
Device_Category	C	High-level equipment group
Device_Type	D	Specific equipment type
Manufacturer	E	Manufacturer label
Model	F	Model code
Serial_Number	G	Serial identifier recorded for the device
Facility_Name	H	Facility associated with the record
City	I	Facility city
Region	J	UK region
Department	K	Department using the equipment
Client_Type	L	Healthcare client category
Purchase_Date	M	Recorded purchase date
Installation_Date	N	Recorded installation date
Warranty_End_Date	O	Recorded warranty expiration date
Contract_Type	P	Service or warranty contract category
Insurance_Coverage	Q	Coverage or reimbursement category label
Unit_Price_GBP	R	Price per unit in GBP
Quantity	S	Number of units represented in the record
Gross_Revenue_GBP	T	Revenue before discount
Discount_GBP	U	Discount applied to the recorded value
Maintenance_Cost_GBP	W	Preventive and corrective maintenance expense
Parts_Cost_GBP	X	Replacement parts expense
Training_Cost_GBP	Y	Training and onboarding expense
Downtime_Hours	AB	Recorded hours of unavailability
Service_Tickets	AC	Number of recorded service tickets
Technician_Name	AD	Assigned engineer or technician
Device_Status	AE	Stored operational status
Compliance_Status	AF	Stored compliance review status
Invoice_Status	AI	Stored payment collection status
Customer_Satisfaction_Score	AJ	Recorded customer feedback score

Source: Data_Dictionary!A5:B51; Raw_Data and Cleaned_Data headers. Descriptions condensed without changing the field meaning.

14 Data dictionary for added fields

Cleaned_Data adds these 16 fields. Their stored values support the summaries, while the classification expressions and date basis require fuller documentation for future refreshes.

Field	Col	Definition
Net_Revenue_GBP	V	Gross revenue less discount
Total_Service_Cost_GBP	Z	Maintenance, parts and training costs combined
Total_Device_Expense_GBP	AA	Gross device value plus service costs
Warranty_Status	AG	Stored warranty classification
Risk_Category	AH	Operational grouping described as based on downtime, tickets and compliance
Installation_Year	AK	Year from installation date
Install_Month	AL	Month label from installation date
Month_Year	AM	Month and year label from installation date
Quarter	AN	Quarter from installation date
Device_Age_Years	AO	Stored device age at the project calculation date
Device_Age_Group	AP	Stored age band
High_Service_Cost_Flag	AQ	High Cost or Normal classification
Downtime_Indicator	AR	Long Downtime or Normal classification
Renewal_Priority	AS	Immediate, High, Medium or Normal priority
Replacement_Recommended	AT	Stored indicator for replacement review
Operational_Improvement_Flag	AU	Needs Review or Stable classification

Source: Cleaned_Data!A1:AU1; Data_Dictionary!A26:B51.

Rules demonstrated by the stored data

The financial relationships can be reproduced directly: gross revenue equals unit price multiplied by quantity, net revenue equals gross revenue less discounts, total service cost equals the three service components, and total device expense equals gross revenue plus total service cost. Date components also follow the installation date.

The observed High Cost group begins at £10,008.10, while the highest Normal service cost is £9,991.27. These values are consistent with a £10,000 cutoff, but they do not establish the exact rule at the boundary. Likewise, the stored downtime groups leave a gap between 42.1 and 53.6 hours, so the precise threshold cannot be proven from labels alone. These examples explain why rule expressions should accompany the derived fields.

The report uses the supplied risk, warranty and priority labels without substituting inferred thresholds. Percentages use the stated denominator, and overlapping groups are reported separately.

Source: Cleaned_Data!Z2:Z3001, AB2:AB3001, AQ2:AR3001. Threshold comments are observations, not confirmed calculation specifications.

15 Evidence references and analytical method

Primary source: Medical_Device_Analytics_Dashboard_Shraddha_Yadav_FINAL_REAL_PIVOT(3).xlsx. Project attribution is recorded as Shraddha Yadav in Project_Overview!B4 and Dashboard!A2. All numerical findings in this report come from this workbook or arithmetic performed on its records.

PivotTable reference index

Analysis	Range on Pivot_Tables	Aggregation
Facility records	A5:B24	Count of Device_ID
Department service cost	A29:B42	Sum of Total_Service_Cost_GBP
Facility satisfaction	A47:B66	Average satisfaction score
City records	A71:B90	Count of Device_ID
Invoice status value	A95:B100	Sum of Net_Revenue_GBP
Category service tickets	A105:B116	Sum of Service_Tickets
Monthly installations	A121:B158	Count of Device_ID
Technician service tickets	D5:E26	Sum of Service_Tickets
Coverage revenue	D31:E37	Sum of Net_Revenue_GBP
Stored age profile	D42:E46	Count of Device_ID
Department downtime	D51:E64	Sum of Downtime_Hours
Compliance status	D69:E73	Count of Device_ID
Risk category	D78:E83	Count of Device_ID

Dashboard chart source index

On Chart_Data, facility counts use A3:B12, department service cost A17:B28, city counts A33:B42, payment value A47:B50, installation-cohort trends A55:C90, device status A95:B98, contract cost A103:B108, risk A113:B116, warranty A121:B123, technician tickets A128:B137, satisfaction A142:B151 and department downtime A156:B167.

Method and verification boundary

The analysis inspected worksheet values, formulas, Excel tables, PivotTable definitions, cache values, chart source references and validation settings. It compared original fields by Device_ID, checked completeness and date order, recomputed row arithmetic and independently grouped the cleaned records. Financial relationships were checked to a half-penny tolerance to accommodate floating-point storage. Report amounts are rounded to two decimals unless a chart explicitly uses millions.

Baseline reconciliation covered six KPI values, 176 PivotTable values and 157 chart-source values. The original workbook was analysed without editing. Interactive refresh and filter behaviour were assessed from the saved formulas and definitions; the report does not claim a desktop Excel refresh test. Source references throughout identify the supporting sheets and cells.